FORM CODE 3X

SECTION 1

Time—30 minutes

45 QUESTIONS

For each question in this section, choose the best answer and blacken the corresponding space on the answer sheet.

Each question below consists of a word in capital letters, followed by five lettered words or phrases. Choose the word or phrase that is most nearly opposite in meaning to the word in capital letters. Since some of the questions require you to distinguish fine shades of meaning, consider all the choices before deciding which is best.

Example:

GOOD: (A) sour (B) bad (C) red (D) hot (E) ugly

Ⓐ ● Ⓒ Ⓓ Ⓔ

1. BRITTLE: (A) dim (B) flexible (C) soluble (D) reliable (E) transparent

2. INHOSPITABLE: (A) sensitive (B) healthy (C) inoffensive (D) enduring (E) welcoming

3. COMPRESS: (A) displace (B) entangle (C) expand (D) soothe (E) delay

4. CONVINCE: (A) dissuade (B) reveal (C) impede (D) betray (E) refuse

5. ONLOOKER: (A) seeker (B) failure (C) fugitive (D) participant (E) thinker

6. SLACKEN: (A) enjoy (B) accumulate (C) accelerate (D) set free (E) turn aside

7. INCOMPATIBLE: (A) receptive (B) meticulous (C) harmonious (D) finished (E) fallible

8. FACILITATE: (A) hamper (B) shorten (C) possess (D) equalize (E) anticipate

9. QUAINT: (A) agreeable (B) brave (C) ordinary (D) comfortable (E) affordable

10. WITTICISM: (A) conclusion (B) insult (C) comment (D) rebuttal (E) platitude

11. FALLOW: (A) humid (B) profound (C) undeviating (D) actively cultivated (E) excessively tedious

12. DEFERENCE: (A) abnormality (B) isolation (C) cowardice (D) disrespect (E) pessimism

13. EMBELLISH: (A) revise (B) abbreviate (C) collaborate (D) examine carefully (E) challenge defiantly

14. EXECRABLE: (A) feeble (B) persistent (C) momentous (D) admirable (E) sensible

15. ATROPHY: (A) growth (B) neutralization (C) ambition (D) selection (E) conformity

Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are five lettered words or sets of words. Choose the word or set of words that best fits the meaning of the sentence as a whole.

Example:

Although its publicity has been ---, the film itself is intelligent, well-acted, handsomely produced, and altogether ----.

(A) tasteless. .respectable (B) extensive. .moderate (C) sophisticated. .amateur (D) risqué. .crude (E) perfect. .spectacular

● Ⓑ Ⓒ Ⓓ Ⓔ

16. John's back ailment was very painful, but he was not ---- and could manage to complete his performance.

(A) inconvenienced (B) incapacitated (C) indifferent (D) competitive (E) aggressive

17. To understand the people of another culture, one must not form a ---- impression; one must play the role of the ---- and evaluate all of the evidence carefully.

(A) distrustful. .skeptic
(B) sympathetic. .radical
(C) hasty. .scientist
(D) graphic. .artist
(E) biased. .guide

18. One reason for optimism is that the social institutions responsible for our present difficulties are not as ---- as they appear to be.

(A) immutable (B) harmless (C) flawless (D) divisible (E) improvable

19. All too often in this book ---- conclusions distort illuminating descriptions and tendentious captions spoil ---- photographs.

(A) illogical. .inane
(B) pertinent. .irrelevant
(C) inevitable. .passable
(D) outlandish. .superfluous
(E) unjustified. .evocative

20. Humor springs from an unexpected and, more specifically, ---- juxtaposition of ideas so that the audience is surprised into laughter.

(A) unnecessary (B) common (C) slapstick (D) incongruous (E) deliberate

GO ON TO THE NEXT PAGE

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1

Each passage below is followed by questions based on its content. Answer all questions following a passage on the basis of what is stated or implied in that passage.

(This passage was published in 1964.)

The issues of poverty and civil rights are deeply intertwined, presenting the central moral problem of our time. Although it is true that three-quarters of the poor are white, it is a tragic fact that half the nonwhites are poor.

The statistics of poverty bear witness to the results of years of discrimination and apathy on the part of the white majority. In 1962, the average income of a white family was $5,642, that of a black family was $3,023, and that of an American Indian family was $1,500. According to the 1960 census figures, Mexican Americans did not fit into a single classification in which their average family income even approximated $3,000.

Contrary to the widely held belief that American Indians receive some sort of consistent dole from the federal government, they are as dependent on the general economy to provide opportunities and jobs as is the rest of the population. Often unable to gain acceptance in the general community, they tend to stay on the reservations, which cannot provide more than minimal employment at low levels. Forty to fifty per cent, seven to eight times the national average, are unemployed.

Of the approximately 4.5 million Mexican Americans, almost one-quarter become migrant workers, following the harvest from state to state. Because they lack a fixed place of residence and employment, migrant workers are neither protected by federal laws nor eligible for public assistance.

Discrimination costs this country between thirteen and seventeen billion dollars every year in revenue and production. This is more than a fourth of the amount spent for national defense. This does not include the incalculable costs we pay in higher crime rates, poor health, mounting welfare costs, and the countless other indirect costs that flow from discriminatory practices.

A great barrier began to swing aside on the day when the Civil Rights Bill passed—a barrier that for generations had been damming tremendous intellectual resources and incalculable energy and vitality, all lost to the American nation.

I say "began." For if the door is being unlocked, if the door is swinging open, it is only ajar. There will be no miracle wrought overnight. Rather, then will come the real test of the maturity of our people. The test will be whether all people in America can work together to dissolve the barriers that have grown and to develop a healthier sense of community. That task will test the wisdom, courage, patience, and judgment of all of us.

21. The passage primarily concerns
    (A) the results of the passage of the Civil Rights Bill
    (B) the economic effects of discrimination
    (C) the task of building a community-oriented America
    (D) the employment practices of impoverished Americans
    (E) practical ways of eliminating the burdens of poverty

22. The author indicates that the passage of the Civil Rights Bill should be regarded as
    (A) an event whose importance has been overemphasized
    (B) a reward for the efforts of all mature Americans
    (C) the first step in a long and difficult, but necessary, process
    (D) a goal that most Americans have worked toward
    (E) the final solution to the dilemmas of poverty and civil rights

23. It can be inferred that the sources of the "energy and vitality" mentioned in lines 44-45 are
    (A) branches of the federal government
    (B) members of the white majority
    (C) politicians who supported the Civil Rights Bill
    (D) Americans whose income is less $3,000
    (E) minority groups that have suffered from discrimination

24. Which of the following best describes the key that unlocks the "door" mentioned by the author in lines 46-48 ?
    (A) The passage of the Civil Rights Bill
    (B) The end of poverty for nonwhite Americans
    (C) The enormous costs of discrimination
    (D) The courage of every American citizen
    (E) Opportunities for new jobs for all Americans

25. The author's intention in the last paragraph of the passage is apparently to
    (A) suggest specific solutions
    (B) warn of unexpected dangers
    (C) criticize exaggerated reports
    (D) inspire constructive attitudes
    (E) soothe agitated tempers

GO ON TO THE NEXT PAGE

The reading passages in this test are brief excerpts or adaptations of excerpts from published material. The ideas contained in them do not necessarily represent the opinions of the College Board or Educational Testing Service. To make the text suitable for testing purposes, we may in some cases have altered the style, contents, or point of view of the original.

The sun has been found surprisingly inconstant both in its rotation and its sunspot activity. Yet it also appears that sunspot occurrences, while irregular, are controlled by a highly precise timer deep within the sun. The intervals between maximum sunspot activity average out to about eleven years, but since magnetic fields on the sun reverse themselves with each cycle, it actually takes about twenty-two years for the sun to return to its original configuration. It has recently been found that the full cycle is controlled by an internal timer that maintains a stable period of 22.27 years, the error margin being only 0.08 year. A possible explanation for the failure of sunspots to adhere to this precise rate may be that the transport of a magnetic field from deep interior to surface is subject to irregularities induced by turbulence in the convective layer, the region in which hot gas rises to the solar surface, cools, and subsides again.

A regularity in climate fluctuations on earth has been attributed to the sunspot cycle. The fluctuations are evident in the varying ratio of deuterium to hydrogen in cellulose extracted from tree rings. The ratio in air moisture seems controlled by atmospheric and sea surface temperature, and evidence of this becomes incorporated into the wood of each annual tree ring. Available data on the formation of tree rings fit the long-term sunspot rhythm, supporting the belief in a link between such activity and the climate.

It is now widely believed that, from 1645 to 1715, virtually no sunspots were evident; the chilling of the climate during this period is known as the "Little Ice Age." The timer deep inside the sun, however, did not miss a beat: when the cycle again became evident, sunspot activity was in step with the earlier rhythm. But it has recently been found that the sun underwent remarkable changes in rotational behavior just before that 70-year intermission, now referred to as the Maunder Minimum. As the minimum and the "Little Ice Age" approached, the rotational velocity of sunspots increased, primarily in the equatorial regions. (The sun, being gaseous, does not rotate as a solid body; normally, the equator rotates every 25 days, the latitudes near the pole every 34.) According to available evidence, the difference between the rotation rate near the equator and that near the poles increased threefold as the minimum approached.

26. According to the author of the passage, the explanation given for the sunspots' failure to adhere to a precise rate of occurrence is

(A) the only credible explanation
(B) a recently documented theory
(C) a highly controversial issue
(D) an unproven hypothesis
(E) a questionable and probably untrue assumption

27. According to the passage, which of the following is (are) true about "the full cycle" (line 11) of sunspot activity?

I. It is repeated approximately once every 22 years.
II. It can be roughly divided into two periods of 11 years.
III. It affects the rotation of the sun.

(A) I only (B) II only (C) I and II only
(D) I and III only (E) I, II, and III

28. It can be inferred from the passage that the word "minimum," as used in "Maunder Minimum" (lines 40-41), probably refers to the decline in the

(A) earth's temperature
(B) number of sunspots
(C) atmospheric ratio of deuterium to hydrogen
(D) relative speed of the sun's rotation
(E) variety of discernible tree rings

29. According to the passage, if in the future the earth were to observe an increase in the rotational velocity of sunspots, we should probably expect which of the following to occur?

I. A general cooling of the earth's climate
II. An increase in the number of sunspots visible on the earth
III. A change in the cycle of the sun's internal timer

(A) I only (B) II only (C) III only
(D) I and II only (E) I and III only

30. The author includes the parenthetical sentence (lines 44-46) for which of the following reasons?

(A) To provide an example supporting a generalization
(B) To state a proposition that is explained in the passage
(C) To present information that clarifies a previous statement
(D) To emphasize the main point of the passage
(E) To shift the description from the abstract to the concrete

GO ON TO THE NEXT PAGE

Select the word or set of words that best completes each of the following sentences.

31. The Mediterranean fruit fly proved to be a ---- to farmers, a pest more ---- to fruits and vegetables than even the Japanese beetle.

(A) delight. .indifferent (B) dilemma. .averse
(C) boon. .deadly (D) triumph. .partial
(E) nightmare. .harmful

32. The assassination of Zapata had ---- the agrarian movement, but had not ---- it; peasant guerrillas still fought, and an Agrarian Party was founded in Mexico City to defend Zapata's ideals.

(A) destroyed. .deterred
(B) impaired. .ended
(C) generated. .decimated
(D) produced. .exterminated
(E) engendered. .endangered

33. Much in keeping with its ---- start, the cooperative is doing a thriving business and is making plans to expand in the near future.

(A) problematic (B) premature
(C) blemished (D) auspicious
(E) erratic

34. This book is particularly valuable because it obliges us to ---- our assumption that source material in Central Africa is too meager to allow more than ---- accounts of local history.

(A) devise. .personal (B) defend. .comprehensive
(C) disclose. .radical (D) limit. .optional
(E) modify. .impressionistic

35. The gaily lit garlands strung across Oxford Street, the enticing shop windows, the crowded theaters—all of these things ---- the ---- of the economic crisis that has come to haunt the country.

(A) bespeak. .severity
(B) underline. .existence
(C) belie. .gravity
(D) protest. .superficiality
(E) malign. .enormity

Each question below consists of a related pair of words or phrases, followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

Example:

YAWN : BOREDOM :: (A) dream : sleep
(B) anger : madness (C) smile : amusement
(D) face : expression (E) impatience : rebellion

Ⓐ Ⓑ ● Ⓓ Ⓔ

36. TEACHER : FACULTY :: (A) actor : cast
(B) student : library (C) painter : mural
(D) composer : orchestra (E) soloist : audience

37. DEVOUR : FOOD :: (A) conserve : energy
(B) borrow : money (C) sweeten : candy
(D) soften : noise (E) guzzle : liquid

38. GHASTLY : HORROR :: (A) soothing : injury
(B) momentary : time (C) indifferent : stupidity
(D) excruciating : agony (E) feeble : power

39. CREST : WAVE :: (A) trunk : tree
(B) shore : lake (C) hub : wheel
(D) base : triangle (E) peak : mountain

40. CHAMELEON : LIZARD :: (A) eagle : hawk
(B) zebra : camel (C) rainbow : sun
(D) blossom : root (E) peacock : bird

41. GULLIBLE : CREDULOUS ::
(A) ravenous : voracious (B) overflowing : thirsty
(C) secretive : curious (D) frivolous : needy
(E) gorgeous : appreciative

42. PERFUNCTORY : ENTHUSIASM ::
(A) persistent : uniformity
(B) tolerant : indecision
(C) impulsive : effectiveness
(D) hesitant : certainty
(E) ambitious : interest

43. NERVE : TEMERITY ::
(A) courage : eminence
(B) skill : finesse
(C) conviction : reliance
(D) foresight : learning
(E) determination : coordination

44. MEDLEY : MUSIC :: (A) panorama : vision
(B) tapestry : thread (C) prism : color
(D) patchwork : cloth (E) encyclopedia : textbook

45. SHRIVELED : TURGIDITY ::
(A) hoary : decadence
(B) prolific : progeny
(C) ambulatory : motion
(D) comatose : animation
(E) ethereal : inspiration

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY.
DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

SECTION 2

Time—30 minutes

25 QUESTIONS

In this section solve each problem, using any available space on the page for scratchwork. Then decide which is the best of the choices given and blacken the corresponding space on the answer sheet.

The following information is for your reference in solving some of the problems.

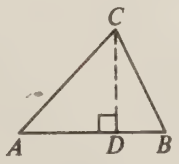
Circle of radius r: Area $= \pi r^2$; Circumference $= 2\pi r$

The number of degrees of arc in a circle is 360.

The measure in degrees of a straight angle is 180.

Definitions of symbols:

$=$	is equal to	$\leqq$	is less than or equal to
$\neq$	is unequal to	$\geqq$	is greater than or equal to
$<$	is less than	$\parallel$	is parallel to
$>$	is greater than	$\perp$	is perpendicular to



Triangle: The sum of the measures in degrees of the angles of a triangle is 180.

If $\angle CDA$ is a right angle, then

(1) area of $\triangle ABC = \dfrac{AB \times CD}{2}$

(2) $AC^2 = AD^2 + DC^2$

Note: Figures which accompany problems in this test are intended to provide information useful in solving the problems. They are drawn as accurately as possible EXCEPT when it is stated in a specific problem that its figure is not drawn to scale. All figures lie in a plane unless otherwise indicated. All numbers used are real numbers.

---

1. If $\dfrac{x - x + x}{2} = \dfrac{1}{2}$, then $x =$

   (A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) 1 (D) 2 (E) 4

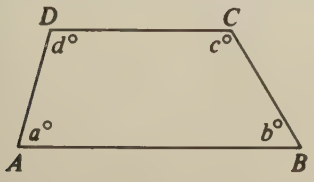
2. If $\frac{3}{5}$ of n is 30, then $\frac{2}{5}$ of n is

   (A) 6
   (B) 10
   (C) 12
   (D) 18
   (E) 20

3. If $x \nearrow y$ means that x^2 is less than y^2, then which of the following relations is true?

   (A) $\frac{1}{3} \nearrow \frac{1}{2}$ (B) $\frac{1}{3} \nearrow \frac{1}{30}$ (C) $2 \nearrow \frac{1}{2}$

   (D) $3 \nearrow 2$ (E) $4 \nearrow 2$

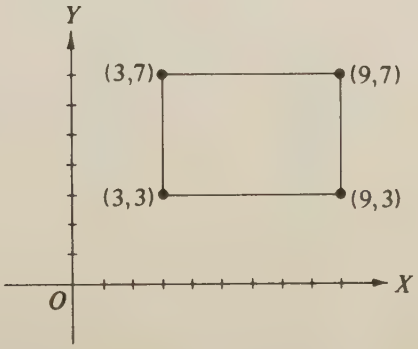


4. In the figure above, AB is parallel to CD. What is the value of $a + b$?

   (A) 90 (B) 120 (C) 150 (D) 180
   (E) It cannot be determined from the information given.

5. If n is a multiple of 3, which of the following is also a multiple of 3 ?

   (A) $n - 2$
   (B) $n + 2$
   (C) $2n + 6$
   (D) $3n - 5$
   (E) $3n + 4$



6. In the figure above, the area of the rectangle is

   (A) 63 (B) 36 (C) 27 (D) 24 (E) 18

GO ON TO THE NEXT PAGE

+ +	+ −	− −	− +
− 0	+ +	+ 0	0 −
0 +	0 0	+ −	− 0

7. In the array of pairs of signs above, a pair has value 1 if a + sign is paired with another + sign or with a 0; any other combination has value 0. What is the total value of the array?

(A) 2 (B) 3 (C) 4 (D) 6 (E) 7

8. $3^4 \cdot 3^2 =$

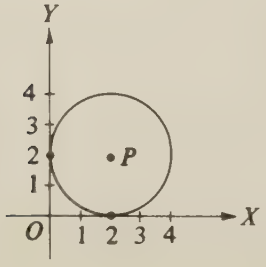
(A) 3^2
(B) 3^6
(C) 3^8
(D) 9^6
(E) 9^8

9. If the perimeter of $\triangle ABC$ is 50 units and if side AB is 2 units longer than side BC, what is the length of side AC ?

(A) 2
(B) 16
(C) 24
(D) 25
(E) It cannot be determined from the information given.

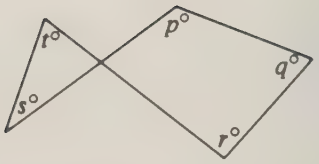
10. Half of the sum of two numbers is 7. If one of the numbers is 3, then the product of the two numbers is

(A) 10 (B) 15 (C) 21 (D) 24 (E) 33



11. If $C(P, r)$ represents a circle with center at the point P and radius r, then the circle in the figure above can be represented as

(A) $C(P, 2)$ (B) $C(P, 1)$ (C) $C(P, 2\sqrt{2})$
(D) $C(P, 4)$ (E) $C(P, 4\pi)$

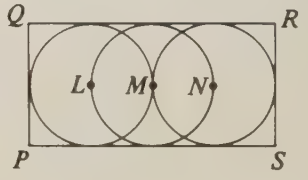


12. Five line segments intersect to form the figure above. If $p + q + r = 285°$, what is the value of $t + s$?

(A) 60
(B) 75
(C) 90
(D) 105
(E) It cannot be determined from the information given.

13. Four empty trucks are on a bridge that has a maximum capacity of 30,000 pounds. If the average (arithmetic mean) weight of three of the trucks is 7,800 pounds, then the greatest possible safe weight of the fourth truck is

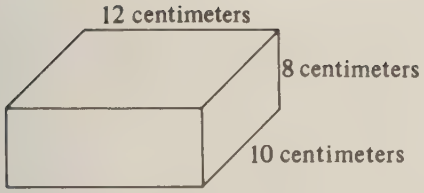
(A) 6,000 lb.
(B) 6,600 lb.
(C) 6,800 lb.
(D) 7,600 lb.
(E) 8,000 lb.



14. In the figure above, circles with centers at L, M, and N each have a radius of 2 and are placed as shown. If $PQRS$ is the smallest rectangle that will enclose the 3 circles, what is the area of $PQRS$?

(A) 8
(B) 12
(C) 16
(D) 32
(E) 48

GO ON TO THE NEXT PAGE



15. How many cubes 2 centimeters on an edge can be packed into the box shown above?

(A) 960 (B) 480 (C) 192
(D) 120 (E) 15

16. If 1 skip equals 3 hops and 1 jump equals 2 skips, how many jumps equal "a hop, a skip, and a jump"?

(A) $1\frac{1}{8}$

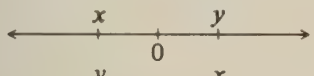
(B) $1\frac{2}{3}$

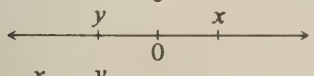
(C) $2\frac{1}{6}$

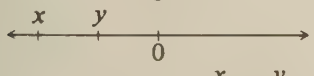
(D) 3

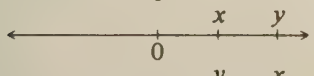
(E) $3\frac{1}{6}$

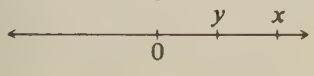
17. If x and y are nonzero integers and if $x^2 + y^3 = 0$ and $x^3 + y^5 = 0$, which of the following number lines shows the relative positions of x, y, and 0 ?

(A) 

(B) 

(C) 

(D) 

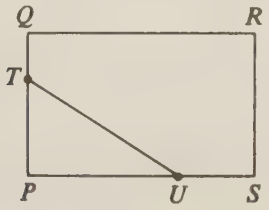
(E) 

18. A stack of red and blue chips is 5 centimeters high. If the red chips are 0.4 centimeters thick and the blue chips are 0.3 centimeters thick, what is the greatest possible number of blue chips that could be in the stack?

(A) 16
(B) 15
(C) 14
(D) 12
(E) 10

19. What is the maximum number of circles that can be made from a piece of string 100 meters long, if the radius of each circle is 5 meters?

(A) 2
(B) 3
(C) 4
(D) 5
(E) 6



20. If, in rectangle $PQRS$ above, $PT = \frac{2}{3}PQ$ and $PU = \frac{2}{3}PS$, what is the ratio of the area of $\triangle PTU$ to the area of $\square PQRS$?

(A) $\frac{2}{9}$ (B) $\frac{1}{3}$ (C) $\frac{4}{9}$ (D) $\frac{2}{3}$

(E) It cannot be determined from the information given.

Population of Yourtown 1900-1950	
Year	Population
1900	500
1910	700
1920	800
1930	1,000
1940	1,300
1950	1,500

21. In the table above, the percent increase in the population was greatest in which of the following periods?

(A) 1900-1910
(B) 1910-1920
(C) 1920-1930
(D) 1930-1940
(E) 1940-1950

GO ON TO THE NEXT PAGE

22. What are all values of b such that $(b - 5)(b + 2)$ is negative?

(A) $-5 < b < 2$
(B) $-2 < b < 5$
(C) $2 < b < 5$
(D) $b < -5$ and $b > 2$
(E) $b < -2$ and $b > 5$

23. If $p + q = 2(r + s)$, which of the following is the average (arithmetic mean) of p, q, r, and s in terms of r and s?

(A) $r + s$ (B) $\frac{3(r+s)}{4}$ (C) $\frac{r+s}{2}$

(D) $\frac{r+s}{4}$ (E) $\frac{r+s}{8}$

24. Rock B in one pan of a scale balances with Rock C plus a 0.6-kilogram weight in the other pan. If the weight of Rock C is 0.6 of B's weight, what is the weight of Rock C?

(A) 0.9 kg
(B) 0.12 kg
(C) 0.36 kg
(D) 1.0 kg
(E) 1.2 kg

25. In how many different color combinations can 3 balls be painted if each ball is painted one color and there are 3 colors available? (Order is not considered; e.g., red, blue, red is considered the same combination as red, red, blue.)

(A) 4 (B) 6 (C) 9 (D) 10 (E) 27

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY.
DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

SECTION 3

Time—30 minutes

40 QUESTIONS

For each question in this section, choose the best answer and blacken the corresponding space on the answer sheet.

Each question below consists of a word in capital letters, followed by five lettered words or phrases. Choose the word or phrase that is most nearly opposite in meaning to the word in capital letters. Since some of the questions require you to distinguish fine shades of meaning, consider all the choices before deciding which is best.

Example:

GOOD: (A) sour (B) bad (C) red (D) hot (E) ugly

Ⓐ ● Ⓒ Ⓓ Ⓔ

1. ACCESSIBLE: (A) impure (B) inactive (C) unimportant (D) purposefully designed (E) difficult to approach

2. INEVITABLE: (A) pleasurable (B) fictitious (C) beneficial (D) avoidable (E) suitable

3. BUNGLE: (A) perform efficiently (B) retire quickly (C) shout angrily (D) energize (E) excuse

4. HYPOTHESIZE:
(A) argue enthusiastically
(B) know with certainty
(C) exaggerate noticeably
(D) plan with care
(E) reflect morally

5. NEGOTIABLE: (A) unpopular (B) illogical (C) unreliable (D) indiscriminate (E) nontransferable

6. REITERATE:
(A) refuse to accept
(B) neglect to examine
(C) fail to repeat
(D) tend to misinterpret
(E) pretend to ignore

7. STEADFASTNESS: (A) lethargy (B) nervousness (C) irresponsibility (D) disorganization (E) irresolution

8. AUTHORITATIVE: (A) partial (B) unscrupulous (C) intelligent (D) submissive (E) fickle

9. INSIPIDNESS:
(A) sweetness and joy
(B) ribaldry and coarseness
(C) indifference and unconcern
(D) spirit and animation
(E) curiosity and acquisitiveness

10. IMPORT: (A) meaninglessness (B) inopportuneness (C) externality (D) negligence (E) humility

Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are five lettered words or sets of words. Choose the word or set of words that best fits the meaning of the sentence as a whole.

Example:

Although its publicity has been ---, the film itself is intelligent, well-acted, handsomely produced, and altogether ---.

(A) tasteless. .respectable (B) extensive. .moderate (C) sophisticated. .amateur (D) risqué. .crude (E) perfect. .spectacular

● Ⓑ Ⓒ Ⓓ Ⓔ

11. The body is protected by a natural defense system that acts in the blood to destroy the army of germs ---- our bodies daily.

(A) guarding (B) vacating (C) attracting (D) invading (E) nurturing

12. Annoyed by his seemingly endless attempts to persuade her to his point of view, Nancy wished that Hamilton were just a bit less ----.

(A) vain (B) controlled (C) persistent (D) impressive (E) devious

13. The scientists reported that the pollution in the coastal areas had already ---- their tentative limit of tolerability and that a higher scale of tolerability would have to be postulated.

(A) restored (B) ignored (C) validated (D) surpassed (E) determined

14. Traditional views assume that violence is somehow ---- "masculine" behavior, rather than a way of behaving which may be ---- by either sex in particular circumstances.

(A) naturally. .adopted
(B) inherently. .perceived
(C) acceptably. .described
(D) accidentally. .selected
(E) avoidably. .intensified

15. Smith contends that psychology is bad poetry ---- science, in that many of its theories are based on unexamined ----, such as the mind as a blank slate, a computer, or a camera.

(A) expanding toward. .hypotheses
(B) inimical to. .facts
(C) escaping from. .quandaries
(D) analagous to. .redundancies
(E) masquerading as. .metaphors

GO ON TO THE NEXT PAGE

3

Each question below consists of a related pair of words or phrases, followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

Example:

YAWN : BOREDOM :: (A) dream : sleep
(B) anger : madness (C) smile : amusement
(D) face : expression (E) impatience : rebellion

Ⓐ Ⓑ ● Ⓓ Ⓔ

16. SEE : INVISIBLE :: (A) correct : inverted
(B) hear : inaudible (C) enjoy : improper
(D) buy : invaluable (E) escape : inaccessible

17. STEAM : GEYSER :: (A) power : generator
(B) atoms : reactor (C) coal : mine
(D) lava : volcano (E) rock : quarry

18. CUSTOMER : SHOPKEEPER :: (A) patron : concert
(B) member : club (C) advocate : policy
(D) peddler : house (E) client : lawyer

19. EPIDERMIS : BODY :: (A) stem : plant
(B) air : diver (C) bark : tree
(D) fur : coat (E) nail : finger

20. OGRE : CRUELTY ::
(A) elf : skepticism
(B) pixie : mischievousness
(C) gnome : punishment
(D) dragon : mythology
(E) goblin : innocence

21. SCRIBBLING : READER :: (A) gibberish : listener
(B) snore : sleeper (C) daub : artist
(D) warble : singer (E) perjury : witness

22. CULMINATION : PROJECT ::
(A) precaution : accident (B) wisdom : judgment
(C) frustration : plan (D) arrival : journey
(E) imitation : idea

23. VERVE : PERFORMANCE :: (A) aspect : feature
(B) valor : courage (C) admiration : talent
(D) imagination : writing (E) sensation : gloom

24. PROMISCUOUS : DISCRIMINATION ::
(A) offensive : aggravation
(B) passive : obedience
(C) impecunious : caution
(D) oblivious : meditation
(E) impulsive : deliberation

25. SATURATE : MOISTEN :: (A) extol : commend
(B) donate : give (C) contaminate : infect
(D) postulate : query (E) assist : hamper

GO ON TO THE NEXT PAGE

Each passage below is followed by questions based on its content. Answer all questions following a passage on the basis of what is stated or implied in that passage.

Above this race of humans stands an immense and tutelary power, which takes upon itself alone to secure their gratification and to watch over their fate. That power is absolute, minute, regular, provident, and mild. It would be like the authority of a parent if, like that authority, its object were to prepare people for adulthood; but it seeks, on the contrary, to keep them in perpetual childhood: it is well content that the people should rejoice, provided they think of nothing but rejoicing. It covers the surface of society with a network of small complicated rules, minute and uniform, through which the most original minds and the most energetic characters cannot penetrate, to rise above the crowd. Such a power does not destroy, but it prevents existence; it does not tyrannize, but it compresses, enervates, extinguishes, and stupefies a people, till each nation is reduced to nothing better than a flock of timid and industrial animals of which the government is shepherd.

26. Which of the following statements summarizes the main idea of the passage?

(A) Creativity finds full expression in a state of anarchy.
(B) Energy not applied toward a specific goal is lost.
(C) Any exercise of power is, by its nature, evil.
(D) Human beings are, by their nature, dependent creatures.
(E) Government is an all-pervasive and inhibiting force.

27. The last sentence suggests that, under the power described, people are kept from all of the following EXCEPT

(A) thinking clearly
(B) acting energetically
(C) working efficiently
(D) creating freely
(E) living fully

28. The author develops the paragraph by means of which of the following?

I. Analogy and metaphor
II. Unsupported generalization
III. Recollected personal experience

(A) I only
(B) I and II only
(C) I and III only
(D) II and III only
(E) I, II, and III

GO ON TO THE NEXT PAGE

3

When Samuel Carboy entered through the swinging door, the floor of the stock exchange was in a state of turmoil. The excitement was due to the still unauthenticated but, alas, well-supported rumor that Bonaparte had beaten the Duke of Wellington. The members to a man were trying to sell. Anyone who could get possession of a chair was standing above the crush and repeating the usual formula with frantic vehemence: "I'm here. I'll sell. D'ye hear me? I'll sell. I've got government securities and I've got bank paper and I've got shipping stocks. I'm not going to give these securities away." One member, Roger Gridsby, varied this by shouting: "Come, men, come! You're Englishmen aren't you? Do you want to take advantage of a temporary darkness on the horizon of England? Do you want to traffic in our monetary despair? Have you no feeling of shame? Support the market, if ye're honest Englishmen. D'ye want to drive prices down to nothing?"

A smartly dressed broker came to Samuel Carboy's side and asked in a cautious tone, "What is it to be today?"

The merchant looked down at him and said, "Westerby, I'm going to buy."

"Buy!" The little broker's voice rose almost to a yelp. "Buy, Samuel? You're joking, Samuel."

"Westerby," declared Carboy, "our army could not be beaten by the usurper Bonaparte. I refuse to believe it. Make what you will out of these rumors. I don't believe them. Westerby, I am going to buy."

"But, Samuel, Samuel!" The broker was speaking now in a beseeching voice. "Haven't you heard about Nathan Longman? He came in here looking like a death's head. That's exactly what he looked like. He said just exactly nothing at all. He just kept stamping his hat firmer on his head and looking very sick. And then he turned and walked out. That means he's selling. If he intended to buy he'd stay here and wait for his prices, wouldn't he? It stands to reason, Samuel."

"Westerby, I'm going to buy."

They retired to a relatively quiet corner and Carboy gave the broker his orders. Westerby marked the prices down on slips of paper, making clucking sounds of disapproval. Then, he shoved the little slips into the silver-embroidered pocket of his plum-colored coat. "It may be some time before they get as low as this," he said. "But they'll get there. You'll be ruined in the end, Samuel, mark my words. If you change your mind, get to me fast."

29. For which of the following reasons was there so much activity on the floor of the exchange?

(A) The army was in retreat.
(B) There was talk of the army's defeat.
(C) The pound had been devalued.
(D) A famous bank had gone bankrupt.
(E) There was fear of a military attack on the exchange.

30. It can be inferred that Nathan Longman is

(A) the owner of a bankrupt business
(B) Carboy's long-standing rival
(C) Carboy's business partner
(D) a friend of the Duke of Wellington
(E) a highly respected trader

31. According to the passage, which of the following were written on the slips of paper by the broker?

(A) Comments on the present worth of the stocks
(B) Names of companies certain to fail
(C) Estimates of the dividends Carboy will earn
(D) The prices Carboy wishes to pay for the stocks
(E) Types of stock owned by Carboy

32. This passage was most likely taken from

(A) a history textbook
(B) a treatise on the effects of hysteria on economics
(C) a historical novel
(D) a biography of the life of Nathan Longman
(E) Samuel Carboy's autobiography

33. The primary purpose of the passage is most probably to present

(A) an explanation of an economic depression
(B) a description of the bartering process used in the stock exchange
(C) an important decision in the life of Samuel Carboy
(D) the role and responsibilities of a broker
(E) the story of Wellington and Bonaparte

GO ON TO THE NEXT PAGE

Since many people are unwilling to accept art for its own sake, they have sought theories that will reduce artistic values to values that are more easily recognized. They seize on doctrines that have been given a limited critical application by serious writers on art and try to make these doctrines the basis of all judgment. Thus we get theories, such as the moralistic view of art in the nineteenth century and the doctrine of functionalism at the present time, which satisfy those who have a craving for explanation but which run counter to the experience of artistically sensitive persons. Art and morals are indeed most closely connected, but the morals of a work of art are to be found less in its ostensible subject matter than in its means of expression. A song may have pious words but a false and wicked tune; a picture may have an improving subject, but the shapes and colors, the very touch of the artist's pencil may be mean and morally revolting. These are truisms, but they were not always present in the mind of even so great a critic as Ruskin and were generally ignored by his contemporaries. The moralistic theory of art suffered when the too crude interpretation of morality that it accompanied was finally brought into popular disfavor. When people ceased to believe that they knew what was good and bad in morals, they were no longer able to discriminate between good and bad in art by the application of moral standards. They had lost faith in everything except material values, values that could be measured. Hence they developed the theory of functionalism, which took its rise from one of several conflicting justifications of Gothic architecture, and began to apply this theory with a kind of desperate simplicity to all the visible arts.

34. The author believes that the connection between art and morality is to be found

(A) primarily in the views of art critics
(B) in the subject matter of a work of art
(C) during periods when people can easily discriminate between good and bad
(D) in the minds of those who are not artistically sensitive
(E) in the form of a work of art more than in its content

35. According to the passage, the moralistic view of art and the theory of functionalism are alike in that both

(A) have been adopted by artistically sensitive persons
(B) are based on a tenet of Gothic architecture
(C) are concerned solely with the means of artistic expression
(D) are justifications of art which go beyond purely artistic values
(E) marked an abrupt change in the moral values of an era

36. The author of the passage implies that most adherents of the moralistic doctrine demanded that a work of art

(A) be moral in both its subject matter and its means of expression
(B) be concerned with some morally improving subject
(C) have no ostensible subject
(D) conform to the standards of artistically sensitive persons
(E) support the material values of the nineteenth century

GO ON TO THE NEXT PAGE

The "sere and yellow leaf" of autumn is likely to bear a tumor. These tumors, called galls, may develop in any part of a plant—leaf, stem, flower, or root—provided the part contains vascular tissue to keep it supplied with nutrients. The derangement of growth and structure may form a tiny wart or a bulging mass the size of a baseball. Such growths are frequently seen in the fall on the withered stalks of goldenrod.

Plant galls inevitably bring to mind their counterpart in animals: the cancer tumor. In support of this analogy, the gall always contains in its interior a mass of amorphous cells. We know, however, that galls are caused by insects or by viruses, molds, and other microorganisms. The parasite supplies some sort of stimulus to initiate abnormal growth, and the galls seem to grow only so long as the stimulus persists. Many animal cancers appear to grow indefinitely without external stimulation. On the other hand, it may turn out that these tumors too are initiated by viruses or by some other infectious process.

Gall growth can originate when the insect causes tissue to develop into new and different cells and promotes more rapid continued growth than would ordinarily occur. In the insect-induced leaf gall, the parasite provokes its host to grow what is in effect a new organ that is adapted to shelter and nurture the insect egg and larva. The amorphous cells within the gall which feed the larva are enclosed in an intricate structure of supporting tissues covered on the outside by epidermal tissue characteristic of the plant stem. From the cellular point of view, a leaf gall bears some similarity to the stem. The globular gall forms as an appendage to the leaf, a kind of fruit in which the insect embryo takes the place of the plant seed. The insect thus exerts a regulatory power over the growth pattern of the plant that suggests a new line of experimental attack on the secrets of cellular specialization and growth.

37. According to the passage, cancerous growths and galls are alike in that both

(A) contain unspecialized cells
(B) appear to grow indefinitely once they have begun
(C) originate under identical conditions
(D) require external stimulation for continuous growth
(E) are modifications of epidermal tissues

38. In the passage, parasite and host refer, respectively, to the

(A) larva and the plant
(B) gall and the insect
(C) larva and the seed
(D) seed and the fruit
(E) gall and the cancer

39. All of the following are necessary for the initial development of a gall EXCEPT

(A) vascular tissue (B) nutrients
(C) epidermal tissue (D) a host
(E) a parasite

40. The author states that a larva in a gall resembles

(A) a tumor in an animal
(B) an insect in an egg
(C) a seed in a fruit
(D) an embryo in an egg
(E) a cell in a stem

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY. DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

SECTION 5

Time—30 minutes

35 QUESTIONS

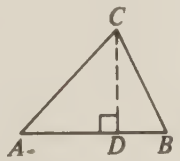
In this section solve each problem, using any available space on the page for scratchwork. Then decide which is the best of the choices given and blacken the corresponding space on the answer sheet.

The following information is for your reference in solving some of the problems.

Circle of radius r: Area $= \pi r^2$; Circumference $= 2\pi r$
The number of degrees of arc in a circle is 360.
The measure in degrees of a straight angle is 180.

Definitions of symbols:

$=$	is equal to	$\leqq$	is less than or equal to
$\neq$	is unequal to	$\geqq$	is greater than or equal to
$<$	is less than	$\parallel$	is parallel to
$>$	is greater than	$\perp$	is perpendicular to



Triangle: The sum of the measures in degrees of the angles of a triangle is 180.
If $\angle CDA$ is a right angle, then

(1) area of $\triangle ABC = \frac{AB \times CD}{2}$

(2) $AC^2 = AD^2 + DC^2$

Note: Figures which accompany problems in this test are intended to provide information useful in solving the problems. They are drawn as accurately as possible EXCEPT when it is stated in a specific problem that its figure is not drawn to scale. All figures lie in a plane unless otherwise indicated. All numbers used are real numbers.

---

1. If $x = 2$ and $y = 98$, then $10(x + y) =$

   (A) 100 (B) 110 (C) 118
   (D) 980 (E) 1,000

2. If x is 40 percent of y and if z is 30 percent of y, then, when $x \neq 0$, the ratio $\frac{z}{x} =$

   (A) 12
   (B) $\frac{5}{6}$
   (C) $\frac{3}{4}$
   (D) $\frac{1}{7}$
   (E) $\frac{1}{12}$

3. If the first two terms in a sequence of numbers are 1 and 2 and each succeeding term is formed by adding all of the preceding terms, what is the fifth term in the sequence?

   (A) 5 (B) 10 (C) 12 (D) 15 (E) 24

4. A month with 5 Wednesdays could start on a

   (A) Sunday
   (B) Monday
   (C) Thursday
   (D) Friday
   (E) Saturday

5. $\frac{2}{3} \cdot \frac{3}{4} \cdot \frac{4}{5} \cdot \frac{5}{6} \cdot \frac{6}{7} \cdot \frac{7}{8} \cdot \frac{8}{9} \cdot \frac{9}{10} \cdot \frac{10}{11} \cdot \frac{11}{12} =$

   (A) $\frac{1}{6}$
   (B) $\frac{11}{36}$
   (C) $\frac{11}{18}$
   (D) $\frac{2}{3}$
   (E) $\frac{11}{12}$



6. In the figure above, the triangle is equilateral and the area of the square is 25. The perimeter of the triangle is

   (A) $\frac{25}{2}$ (B) 15 (C) 20 (D) 25 (E) 75

7. If the average (arithmetic mean) of 3, 4, 5, and x is 6, then $x =$

   (A) 0 (B) 6 (C) 12 (D) 24 (E) 60

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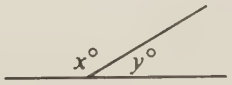
5

Questions 8-27 each consist of two quantities, one in Column A and one in Column B. You are to compare the two quantities and on the answer sheet blacken space

A if the quantity in Column A is greater;
B if the quantity in Column B is greater;
C if the two quantities are equal;
D if the relationship cannot be determined from the information given.

Notes: 1. In certain questions, information concerning one or both of the quantities to be compared is centered above the two columns.
2. In a given question, a symbol that appears in both columns represents the same thing in Column A as it does in Column B.
3. Letters such as x, n, and k stand for real numbers.

EXAMPLES

	Column A	Column B	Answers
E1.	2×6	$2 + 6$	● Ⓑ Ⓒ Ⓓ
E2.	 $180 - x$	y	Ⓐ Ⓑ ● Ⓓ
E3.	$p - q$	$q - p$	Ⓐ Ⓑ Ⓒ ●

Column A Column B

$x = y$
$z > y$

8. x z

An airplane flying at an altitude of 34,000 feet above sea level descends 7,500 feet.

9. The altitude above sea level of the airplane after making this descent — 26,000 feet

---

$x + y = 3$
$x + 2 = -3$

10. x y

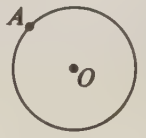
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11. $\frac{78{,}452}{1{,}000}$ 100

$2x + 12 = 10$

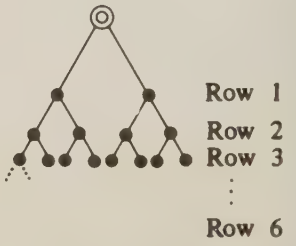
12. $4x + 24$ 20

Column A Column B



P is a point, not shown, inside the circle with center O.

13. AO AP



In each row a black dot has two lines connecting it to two dots in the row beneath.

14. Total number of black dots that would be in row 6 — 60

---

15. The number of multiples of 2 between integers x and $x + 78$ — The number of multiples of 6 between integers x and $x + 78$

Box A has volume 24.
Box B has volume 15.

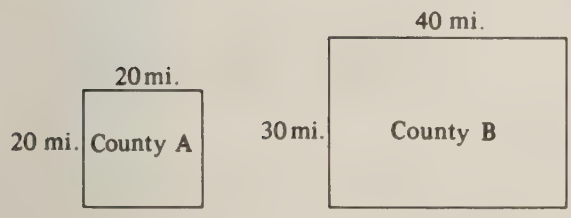
16. Base area of A Base area of B

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SUMMARY DIRECTIONS FOR COMPARISON QUESTIONS

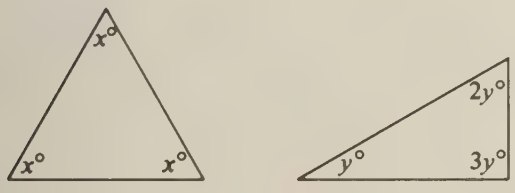
Answer: A if the quantity in Column A is greater;
B if the quantity in Column B is greater;
C if the two quantities are equal;
D if the relationship cannot be determined from the information given.

Column A	Column B



Counties A and B are rectangular.

17. Population of County A if there are 10 people per square mile | Population of County B if there are 5 people per square mile



18. x | $2y$

19. $\frac{0.35}{2}$ | $\frac{1}{2} \times \frac{1}{3}$

Tom is now 5 years older than Bill was 3 years ago. Bill is x years old now.

20. $x + 8$ | Tom's age now in years

In a certain game, there are only four types of moves. These moves advance a playing piece 2, 3, 7, or 9 spaces, respectively, in any order.

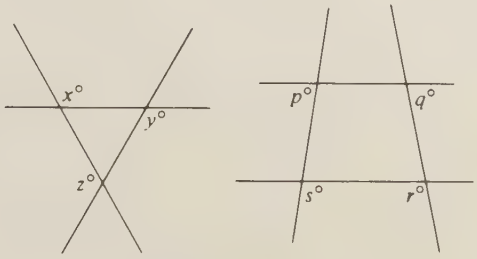
21. The minimum number of moves required to advance a piece exactly 26 spaces | 4

$$2 + y = 6 + y - 4$$

22. y | 2

23. $\sqrt{x^4 + 4x^2 + 4}$ | $x^2 + 2$

Column A	Column B



24. $x + y + z$ | $p + q + r + s$

$n < 1$
$n \neq 0$

25. $\frac{1}{n}$ | n

26. $\frac{x^3}{3}$ | $\left(\frac{x}{3}\right)^3$

MACHINE-PART PRODUCTION
July 1

Company	Number Not Defective	Number Defective	Defective as Percent of Total
A	95	5	5%
B	72	x	10%
C	392	y	2%

27. x | y

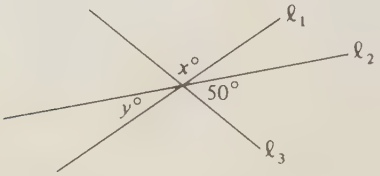
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Solve each of the remaining problems in this section using any available space for scratchwork. Then decide which is the best of the choices given and blacken the corresponding space on the answer sheet.

28. If $x > y$ and if $w > z$, which of the following would establish that $y > z$?

I. $x > w$
II. $y = w$
III. $y > w$

(A) I only (B) II only (C) III only
(D) I and II (E) II and III



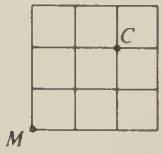
29. In the figure above, three lines intersect in a point. What is the value of x in terms of y ?

(A) $130 - y$ (B) $230 - y$ (C) $260 - 2y$
(D) $310 - 2y$ (E) $50 + y$

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30. If two odd integers are prime, which of the following statements must be true?

(A) Their product is an odd integer.
(B) Their sum is prime.
(C) Their sum is an odd integer.
(D) Their product is prime.
(E) The sum of their squares is prime.



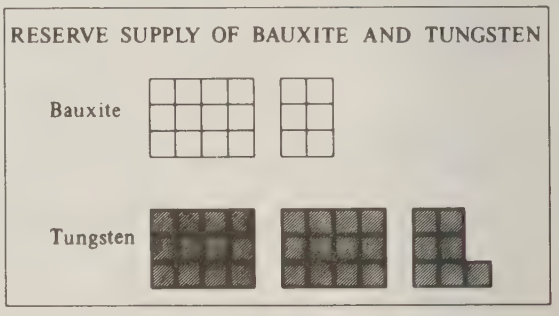
31. A mouse located at point M wants to eat the cheese located at point C. If the mouse can go only up or to the right along the given lines, how many different paths can he take from M to C ?

(A) 12 (B) 10 (C) 8 (D) 6 (E) 4

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32. The sum of two positive consecutive integers is x. In terms of x, what is the value of the smaller of these two integers?

(A) $\frac{x}{2} - 1$ (B) $\frac{x-1}{2}$ (C) $\frac{x}{2}$
(D) $\frac{x+1}{2}$ (E) $\frac{x}{2} + 1$



33. In the chart above, the amount represented by each shaded square is twice that represented by each unshaded square. According to the chart, the amount of tungsten in reserve is how many times greater than the amount of bauxite in reserve?

(A) $\frac{9}{31}$ (B) $\frac{31}{36}$ (C) $\frac{36}{31}$ (D) $\frac{11}{6}$ (E) $\frac{31}{9}$

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34. A restaurant manager notices that t orders of iced tea, each sold at c cents, are served per hour. If the restaurant is open h hours a day, what is the amount of money in dollars received in 1 day from the sale of iced tea?

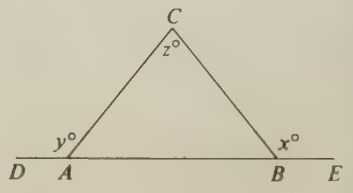
(A) $\frac{100tc}{h}$

(B) tch

(C) $\frac{ch}{t}$

(D) $\frac{tch}{100}$

(E) $\frac{tc}{100h}$



35. In $\triangle ABC$ above, $AC = BC$, and DE is a line segment. If x is 15 less than $2z$, then $y + z =$

(A) 70 (B) 125 (C) 150
(D) 180 (E) 195

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY.
DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

Correct Answers for Scholastic Aptitude Test
Form Code 3X

VERBAL		MATHEMATICAL	
Section 1	Section 3	Section 2	Section 5
1. B	1. E	1. C	1. E
2. E	2. D	2. E	2. C
3. C	3. A	3. A	3. C
4. A	4. B	4. E	4. B
5. D	5. E	5. C	5. A
6. C	6. C	6. D	6. B
7. C	7. E	7. C	7. C
8. A	8. D	8. B	*8. B
9. C	9. D	9. E	*9. A
10. E	10. A	10. E	*10. B
11. D	11. D	11. A	*11. B
12. D	12. C	12. D	*12. C
13. B	13. D	13. B	*13. D
14. D	14. A	14. D	*14. A
15. A	15. E	15. D	*15. A
16. B	16. B	16. B	*16. D
17. C	17. D	17. B	*17. B
18. A	18. E	18. C	*18. C
19. E	19. C	19. B	*19. A
20. D	20. B	20. A	*20. A
21. B	21. A	21. A	*21. C
22. C	22. D	22. B	*22. D
23. E	23. D	23. B	*23. C
24. A	24. E	24. A	*24. C
25. D	25. A	25. D	*25. D
26. D	26. E		*26. D
27. C	27. C		*27. C
28. B	28. B		28. E
29. A	29. B		29. A
30. C	30. E		30. A
31. E	31. D		31. D
32. B	32. C		32. B
33. D	33. C		33. E
34. E	34. E		34. D
35. C	35. D		35. E
36. A	36. B		
37. E	37. A		
38. D	38. A		
39. E	39. C		
40. E	40. C		
41. A			
42. D			
43. B			
44. D			
45. D			

*Indicates four-choice questions. (All of the other questions are five-choice.)

The Scoring Process

Machine-scoring is done in three steps:

- *Scanning.* Your answer sheet is "read" by a scanning machine and the oval you filled in for each question is recorded on a computer tape.
- *Scoring.* The computer compares the oval filled in for each question with the correct response. Each correct answer receives one point; omitted questions do not count toward your score. For each wrong answer, a fraction of a point is subtracted to correct for random guessing. For questions with five answer choices, one-fourth of a point is subtracted for each wrong response; for questions with four answer choices, one-third of a point is subtracted for each wrong response. The SAT-verbal test has 85 questions with five answer choices each. If, for example, a student has 44 right, 32 wrong, and 9 omitted, the resulting raw score is determined as follows:

  $$44 \text{ right} - \frac{32 \text{ wrong}}{4} = 44 - 8 = 36 \text{ raw score points}$$

  Obtaining raw scores frequently involves the rounding of fractional numbers to the nearest whole number. For example, a raw score of 36.25 is rounded to 36, the nearest whole number. A raw score of 36.50 is rounded upward to 37.
- *Converting to reported scaled score.* Raw test scores are then placed on the College Board scale of 200 to 800 through a process that adjusts scores to account for minor differences in difficulty among different editions of the test. This process, known as equating, is performed so that a student's reported score is not affected by the edition of the test taken nor by the abilities of the group with whom the student takes the test. As a result of placing SAT scores on the College Board scale, scores earned by students at different times can be compared. For example, an SAT-verbal score of 400 on a test taken at one administration indicates the same level of developed verbal ability as a 400 score obtained on a different edition of the test taken at another time.

How to Score the Test

You can verify the College Board SAT scores reported to you recently by using the information in this booklet along with the copy of your answer sheet. *Before you begin, check that the first two characters (number and letter) of the form code you marked in item 3 on your answer sheet are the same as the form code printed on the front of this booklet.* Compare the responses shown on the copy of your answer sheet with the list of correct answers.

SAT-Verbal Sections 1 and 3

Step A: Count the number of correct answers for *section 1* and record the number in the space provided on the worksheet on the next page. Then do the same for the incorrect answers. (Do not count omitted answers.) To determine subtotal A, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal A}$$

Step B: Count the number of correct answers and the number of incorrect answers for *section 3* and record the numbers in the spaces provided on the worksheet. To determine subtotal B, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal B}$$

Step C: To obtain C, add subtotal A to subtotal B, keeping any decimals. Enter the resulting figure on the worksheet.

Step D: To obtain D, your raw verbal score, round C to the nearest whole number. (For example, any number from 44.50 to 45.49 rounds to 45.) Enter the resulting figure on the worksheet.

Step E: To find your reported SAT-verbal score, look up the total raw verbal score you obtained in step D in the conversion table on the back cover. Enter this figure on the worksheet. (The SAT-verbal score you just recorded and your reported SAT-verbal score should be identical. If not, see the paragraph at the bottom of the next page.)

SAT-Mathematical Sections 2 and 5

Step A: Count the number of correct answers and the number of incorrect answers for *section 2* and record the numbers in the spaces provided on the worksheet. To determine the subtotal A, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal A}$$

Step B: Count the number of correct answers and the number of incorrect answers for the *five-choice questions (questions 1 through 7 and 28 through 35) in section 5* and record the numbers in the spaces provided on the worksheet. To determine the subtotal B, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal B}$$

Step C: Count the number of correct answers and the number of incorrect answers for the *four-choice questions (questions 8 through 27) in section 5* and record the numbers in the spaces provided on the worksheet. To determine the subtotal C, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{3} = \text{subtotal C}$$

Step D: To obtain D, add subtotal A, subtotal B, and subtotal C, keeping any decimals. Enter the resulting figure on the worksheet.

Step E: To obtain E, your raw mathematical score, round D to the nearest whole number. (For example, any number from 44.50 to 45.49 rounds to 45.) Enter the resulting figure on the worksheet.

Step F: To find your reported SAT-mathematical score, look up the total raw mathematical score you obtained in E in the conversion table on the back cover. Enter this figure on the worksheet. (The SAT-mathematical score you just recorded and your reported SAT-mathematical score should be identical. If not, see the paragraph at the bottom of the next page.)

SAT-SCORING WORKSHEET

FORM CODE 3X

SAT-Verbal Sections

A. Section 1: _____ (no. correct) − ¼ (_____) (no. incorrect) = _____ subtotal A

B. Section 3: _____ (no. correct) − ¼ (_____) (no. incorrect) = _____ subtotal B

C. Total unrounded raw score (Total A + B) _____ C

D. Total rounded raw score (Rounded to nearest whole number) _____ D

E. SAT-verbal reported scaled score (See the conversion table on the back cover.) _____ SAT-verbal score

SAT-Mathematical Sections

A. Section 2: _____ (no. correct) − ¼ (_____) (no. incorrect) = _____ subtotal A

B. Section 5: Questions 1 through 7 and 28 through 35 (5-choice) _____ (no. correct) − ¼ (_____) (no. incorrect) = _____ subtotal B

C. Section 5: Questions 8 through 27 (4-choice) _____ (no. correct) − ⅓ (_____) (no. incorrect) = _____ subtotal C

D. Total unrounded raw score (Total A + B + C) _____ D

E. Total rounded raw score (Rounded to nearest whole number) _____ E

F. SAT-mathematical reported scaled score (See the conversion table on the back cover.) _____ SAT-math score

Should you have any questions on these scoring instructions, you may call the phone number below. If, after following the above scoring directions and checking your work at least twice, your results disagree with the SAT-verbal or SAT-mathematical score reported on your ATP score report, you may request rescoring of your answer sheet. If rescoring confirms that an error had been made (resulting in either higher or lower scores than those originally reported), corrected reports will be sent to all recipients of your original scores. Please send your request to:

College Board ATP
Box 592
Princeton, NJ 08541
Attention: Rescore Request
Telephone: (609) 771-7600

Please indicate whether it is your SAT-verbal or SAT-mathematical score, or both, that you want to be rescored. When you write, please include a copy of this scoring worksheet on which you did your calculations.

Score Conversion Table
Scholastic Aptitude Test
Form Code 3X

Raw Score	College Board Reported Score		Raw Score	College Board Reported Score	
	SAT-Verbal	SAT-Math		SAT-Verbal	SAT-Math
85	800		40	440	610
84	780		39	440	600
83	760		38	430	590
82	750		37	420	580
81	740		36	420	570
80	730		35	410	560
79	720		34	400	560
78	710		33	400	550
77	700		32	390	540
76	690		31	380	530
75	680		30	380	520
74	670		29	370	510
73	660		28	360	500
72	650		27	360	490
71	640		26	350	480
70	640		25	340	480
69	630		24	340	470
68	620		23	330	460
67	620		22	320	450
66	610		21	320	440
65	610		20	310	430
64	600		19	300	420
63	590		18	300	410
62	590		17	290	410
61	580		16	280	400
60	570	800	15	280	390
59	570	780	14	270	380
58	560	770	13	260	370
57	550	760	12	260	360
56	550	750	11	250	350
55	540	740	10	250	340
54	530	730	9	240	330
53	530	720	8	230	330
52	520	720	7	230	320
51	510	710	6	220	310
50	510	700	5	210	300
49	500	690	4	210	290
48	490	680	3	200	280
47	490	670	2	200	270
46	480	660	1	200	260
45	470	650	0	200	250
44	470	640	– 1	200	250
43	460	640	– 2	200	240
42	450	630	– 3	200	230
41	450	620	– 4	200	220
			– 5	200	210
			– 6 or below	200	200